Series & Binomial Theorem

ClipSAT Printable Practice Suite (clipsat.org) • Topic 10.01

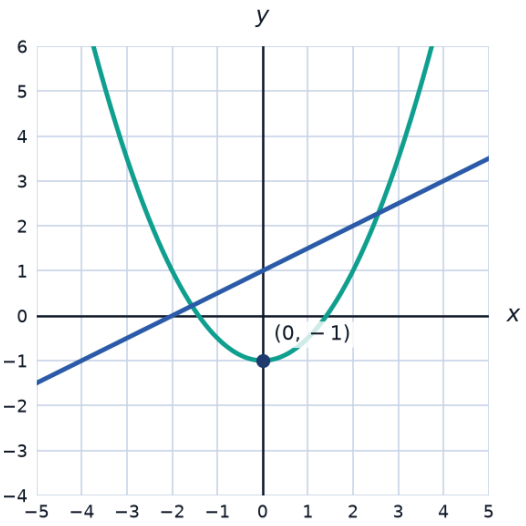
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows the graph of the function $y = f(x)$. The graph is transformed by a sequence of operations. The resulting graph, $y = g(x)$, is also shown. The function $f(x)$ is given by $f(x) = ax^2 + bx + c$, where $a, b,$ and c are constants. The vertex of the graph of $y = f(x)$ is at $(2, -1)$, and it passes through the point $(4, 3)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ by $(-1\ 3)$ followed by a reflection in the y-axis. Find the values of $a, b,$ and c.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^nC_r = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed?

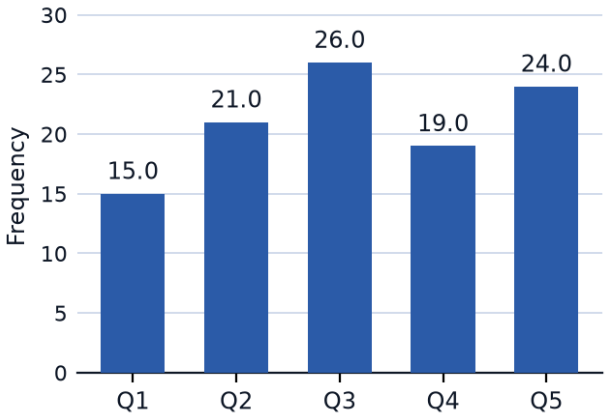
4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A particle of mass 3 kg is at rest on a smooth horizontal surface. A horizontal force of magnitude P N is applied to the particle, causing it to accelerate at a rate of 4 m/s^2. Find the value of P.

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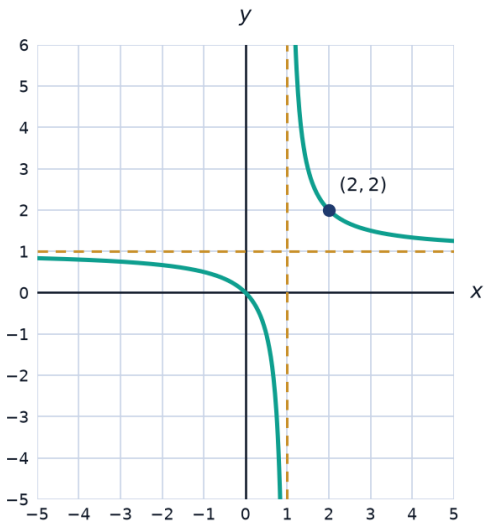
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The two-way table shows the results of a survey of 100 students on their preferred mode of transport to school and their year group. | | Car | Bus | Walk | Total | |----------|-----|-----|------|-------| | Year 1 | 15 | 20 | 10 | 45 | | Year 2 | 25 | 18 | 12 | 55 | | Total | 40 | 38 | 22 | 100 | What is the probability that a randomly selected student is a Year 1 student who prefers to travel by car?



6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A 3 kg particle on a rough surface, $\mu = 0.4$. Horizontal force F N. (a) Find the limiting friction. (b) Acceleration when $F = 20$ N. (c) Time to travel 5 m from rest.



7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A factory tests 30 independent components, each with a fixed probability of being defective; X is the number of defective components. Is X binomially distributed?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

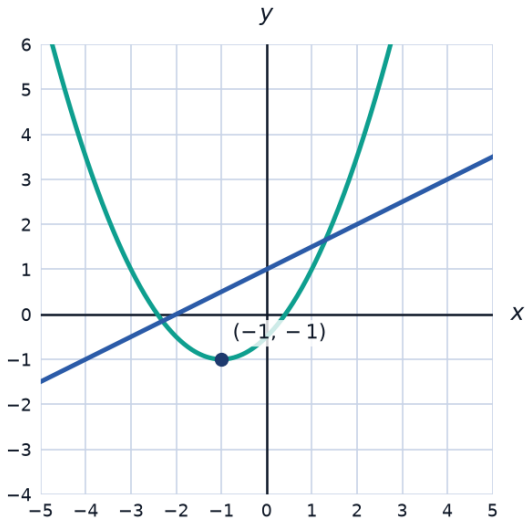
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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A survey asks 50 randomly and independently selected people the same yes/no question; X is the number who say yes. Is X binomially distributed?

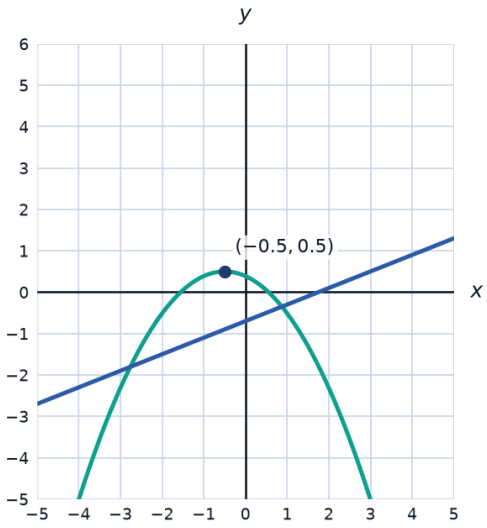
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A machine has a useful power output of 883 W and wastes 367 W as heat. Find its efficiency as a percentage, correct to 1 decimal place.



10 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A particle of mass 5 kg slides down a rough plane inclined at 30° to the horizontal, experiencing a constant friction force of 0 N. Taking $g = 9.8$ m/s^2, find its acceleration down the slope.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Given $X \sim N(\mu, \sigma^2)$, $P(X < 99) = 0.8$ and $P(X < 83) = 0.2$, find μ, correct to the nearest integer.

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12 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A particle of mass m kg is placed on a rough inclined plane. The plane is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$. The particle is on the point of sliding down the plane. The coefficient of static friction between the particle and the plane is μ_s. Given that the acceleration due to gravity is $g = 9.8$ m/s^2, find the value of μ_s.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In an arithmetic progression, the 3rd term is 14 and the 7th term is 34. (a) Find the first term a and the common difference d. (b) Find the sum of the first 25 terms.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Series & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

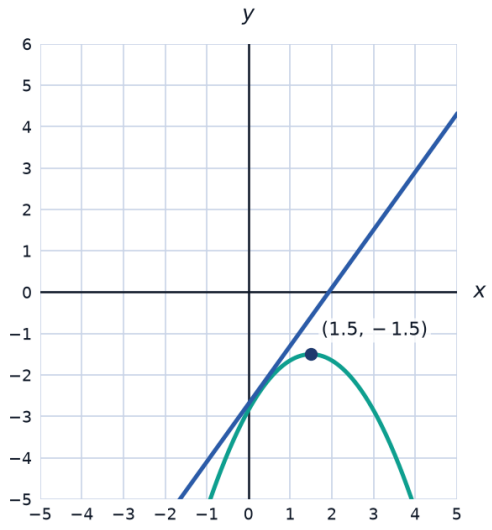
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

In a manufacturing batch, 5% of components are defective. A sample of 12 components is inspected. Find the probability that at least 2 components are defective.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Functions: Composite & Inverse

A fair coin is tossed 10 times. Find the probability of getting (a) exactly 6 heads, (b) fewer than 3 heads.



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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Circular Measure

[Binomial Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Kinematics

Practice 4.1: A multiple-choice examination consists of 20 independent questions, each having 4 options of which only one is correct. A student guesses randomly on all questions. Let X be the number of correct answers. (a) State the distribution of X including its parameters. (b) Find the probability that the student passes by scoring at least 10 correct answers. (c) Calculate the mean and standard deviation of X.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 12.01 Forces & Equilibrium

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An arithmetic progression has first term $a = 7$ and common difference $d = 4$. Find the 20th term u_{20}.

a. 83

b. 87

c. 79

d. 80

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the sum of the first 25 terms of the AP: $3, 8, 13, 18, \ldots$

a. 1575

b. 1625

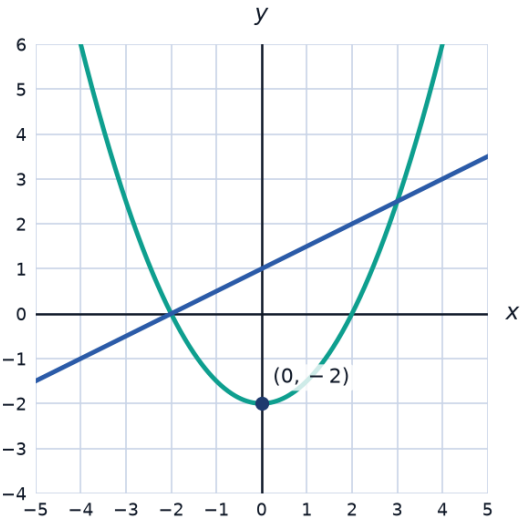
c. 1500

d. 1550

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In an AP, the 5th term is 18 and the 12th term is 39. Find the first term a and common difference d.



a. $a = 6, d = 3$

b. $a = 3, d = 3$

c. $a = 9, d = 3$

d. $a = 6, d = 4$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A geometric progression has first term $a = 5$ and common ratio $r = 2$. Find the 8th term u_8.

a. 640

b. 1280

c. 320

d. 512

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the sum of the first 6 terms of the GP: $3, -6, 12, -24, \ldots$

a. -63

b. 63

c. -127

d. -65

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The 2nd term of a GP is 12 and the 5th term is 324. Find the common ratio r.

a. 3

b. 2

c. 4

d. $\sqrt{3}$

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

State the condition for the sum to infinity of a geometric progression to exist.

a. $|r| < 1$

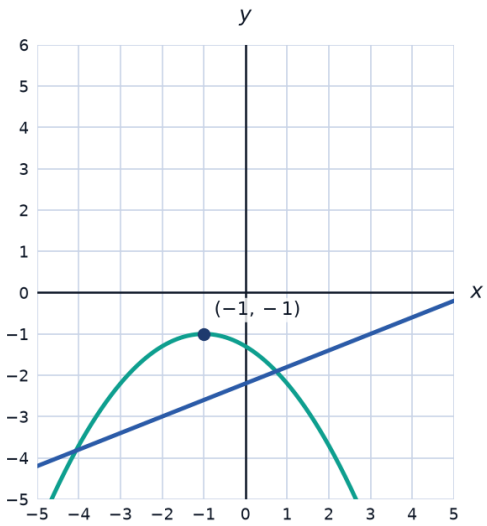
b. $r > 1$

c. $r \leq 1$

d. $|r| > 1$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the sum to infinity of the GP: $16, 12, 9, 6.75, \ldots$



a. 64

b. 48

c. 32

d. 72

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Express the recurring decimal $0.\overline{27} = 0.272727\ldots$ as an exact fraction in lowest terms using an infinite geometric series.

a. $\frac{3}{11}$

b. $\frac{27}{100}$

c. $\frac{9}{33}$

d. $\frac{7}{25}$

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the coefficient of x^3 in the expansion of $(2 + 3x)^5$.

a. 1080

b. 720

c. 540

d. 1440

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

a. 240

b. 60

c. 160

d. 120

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the expansion of $(1 + ax)^6$, the coefficient of x^2 is 60. Given $a > 0$, find a.

a. 2

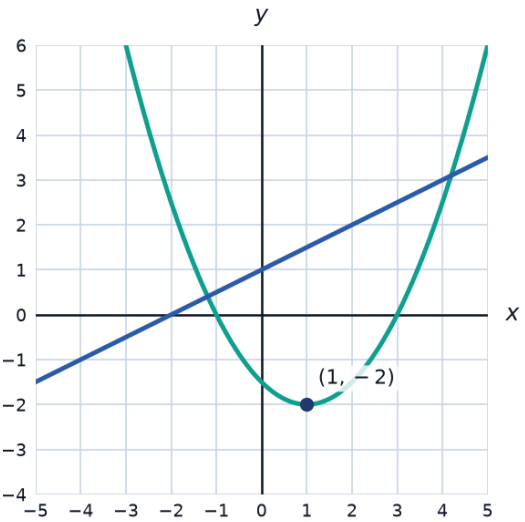
b. 4

c. 3

d. 5

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A geometric series has first term $a = -10$ and common ratio $r = 1/2$. Find S_6, the sum of the first 6 terms.



a. -315/16

b. -155/8

c. -475/16

d. -635/32

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34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the sum of the first 40 terms of an arithmetic sequence with first term -11 and common difference -2.

a. -2000

b. -2004

c. -1994

d. -2003

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

An arithmetic sequence has first term $a = 7$ and common difference $d = 3$. Find the 7th term.

a. 21

b. 25

c. 24

d. 26

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The 4th term of an arithmetic sequence is 19, and the common difference is 3. Find the first term.

a. 8

b. 9

c. 15

d. 10

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An arithmetic sequence has first term 6 and common difference 5. Find n such that $S_n = 1403$.

a. 19

b. 25

c. 27

d. 23

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A geometric series has first term $a = 11$ and common ratio $r = 1/2$. Find S_∞.

a. 11/2

b. 22

c. 22/3

d. 33

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Kinematics

The 12th term of an arithmetic sequence is 58, and the first term is 14. Find the common difference.

a. 4

b. 1

c. 6

d. -1

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 12.01 Forces & Equilibrium

A geometric sequence has first term $a = 8$ and common ratio $r = 1/3$. Find the 4th term.

a. -8/27

b. 8/27

c. 8/81

d. 8/9

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 13.01 Newton's Laws

A factory produces light bulbs. The probability that a randomly selected light bulb is defective is p. The number of defective light bulbs in a random sample of n bulbs follows a binomial distribution $X \sim B(n, p)$. It is known that the mean number of defective bulbs in a sample of $n = 25$ is 2. \\ (i) Calculate the value of p. \\ (ii) For a sample of $n = 50$ bulbs, calculate the probability of finding exactly k defective bulbs, where k is the smallest integer such that $P(X \leq k) > 0.9$. \\ (iii) If a sample of $n = 50$ bulbs is taken, find the smallest value of n' such that the probability of observing at least one defective bulb is greater than 0.999 .

a. 0.04

b. 0.08

c. 0.12

d. 0.16

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 14.01 Energy, Work & Power

A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is $p = \frac{1}{6}$. The number of trials is $n = 4$. This scenario can be modelled by a binomial distribution, denoted as $X \sim B(n, p)$. Which of the following represents the correct parameters for this binomial distribution?

a. $B(4, 1/6)$

b. $B(6, 1/4)$

c. $B(1, 4/6)$

d. $B(4, 4/6)$

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 15.01 Representation of Data

A die is rolled 15 times; X is the number of sixes. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. Yes — but only if $n > 30$

c. Yes — fixed, independent trials with constant success probability

d. No — the number of trials is not fixed

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 16.01 Permutations & Combinations

A fair coin is tossed 20 times; X is the number of heads. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. Yes — but only if $n > 30$

c. Yes — fixed, independent trials with constant success probability

d. No — the number of trials is not fixed

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 17.01 Discrete Random Variables

Cards are dealt from a single shuffled deck without replacement; X is the number of aces dealt. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. Yes — but only if $n > 30$

c. Yes — fixed, independent trials with constant success probability

d. No — the number of trials is not fixed

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